

UEI627: DATA ANALYTICS

L	T	P	Cr
3	0	0	3.0

Course Objectives: To understand the various concepts involved in data analytics.

Introduction: Overview of Data analytics, Descriptive Statistics - Graphical Approaches, Measures of central Tendency, Measures of Dispersion, Random Variables and Probability Distributions.

Inferential Statistics: Single sample tests, two Sample Tests, Type 1 and Type 2 Errors, Confidence Intervals, ANOVA and Test of Independence, Modern Data ecosystem, analytics vs data analysis, Identifying data for analysis, Introduction to Regression, Least Squares Regression, Simple and Multiple Regression in Excel and Matlab, Regularization/Coefficient shrinkage, Data Modelling Approaches and Algorithmic Modelling Approaches. Machine learning with big data

Supervised Learning with regression and classification techniques: Logistic Regression, Training a Logistic Regression Classifier, Classification and Regression Trees, Bias Variance Dichotomy, Model Assessment and Selection.

Unsupervised Learning and Challenges for Big Data Analytics: Clustering, Associative Rule Mining, Challenges for big data analytics, Data Wrangling, Data cleaning and data visualization, Statistical Analysis in mining and visualizing data

Prescriptive analytics: Introduction to Experimentation and Active Learning, Reinforcement Learning.

Course Learning Outcomes (CLO):

Student will be able to:

1. Elucidate the basics of Data Analytics
2. Analyze different techniques of Inferential Statistics
3. Apply various regression techniques for modelling
4. Exhibit the understanding of techniques for data clustering and reinforcement learning.

Text Books:

3. T. Hastie, R. Tibshirani, J. Friedman. *The Elements of Statistical Learning*, 2e, 2008.
4. Christopher Bishop. *Pattern Recognition and Machine Learning*. 2e.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST+EST	70
2	Sessional (May include Assignments//Quizzes/Lab Evaluations)	30